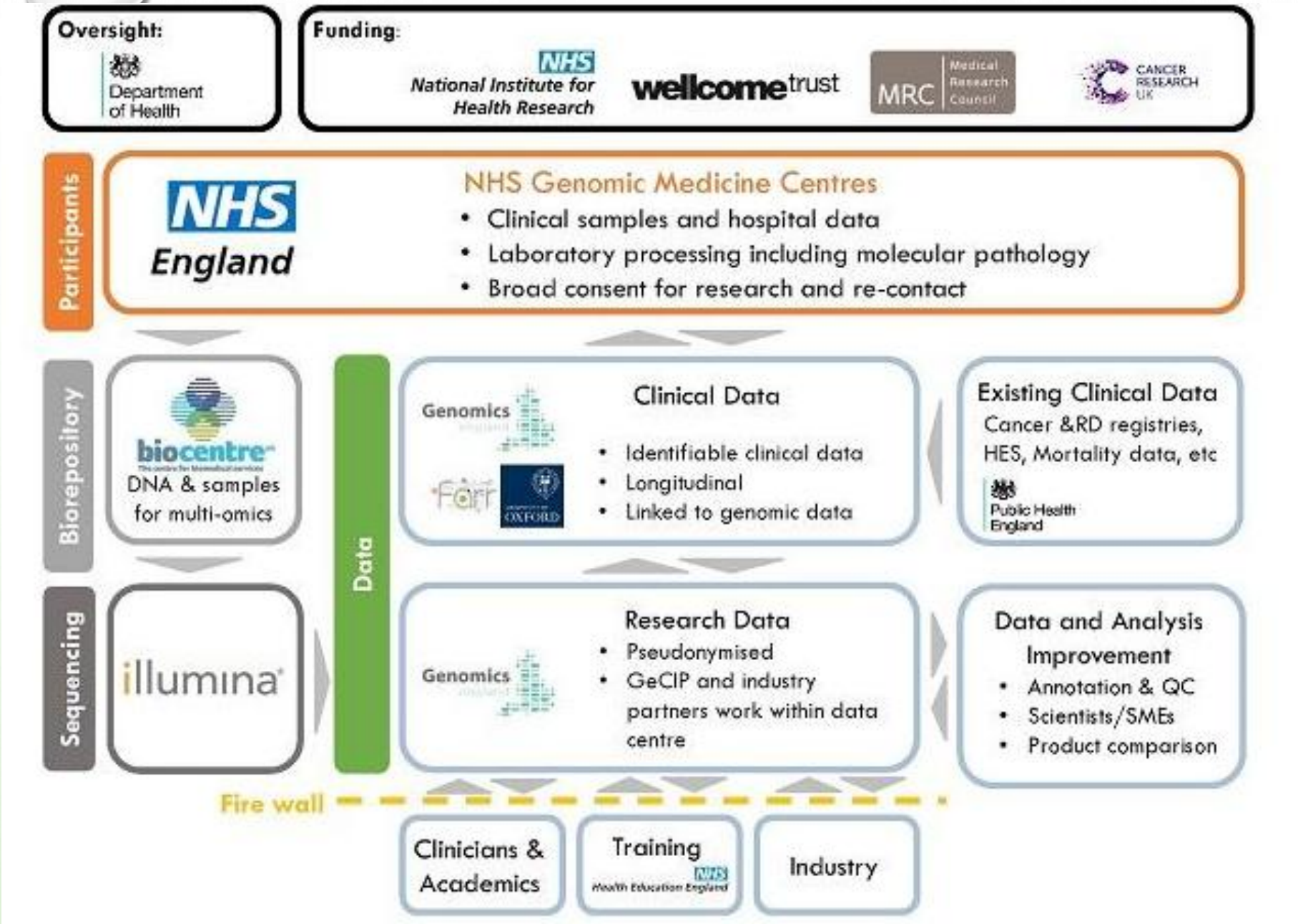
AI LAB 2026

1.2 – THE DATA

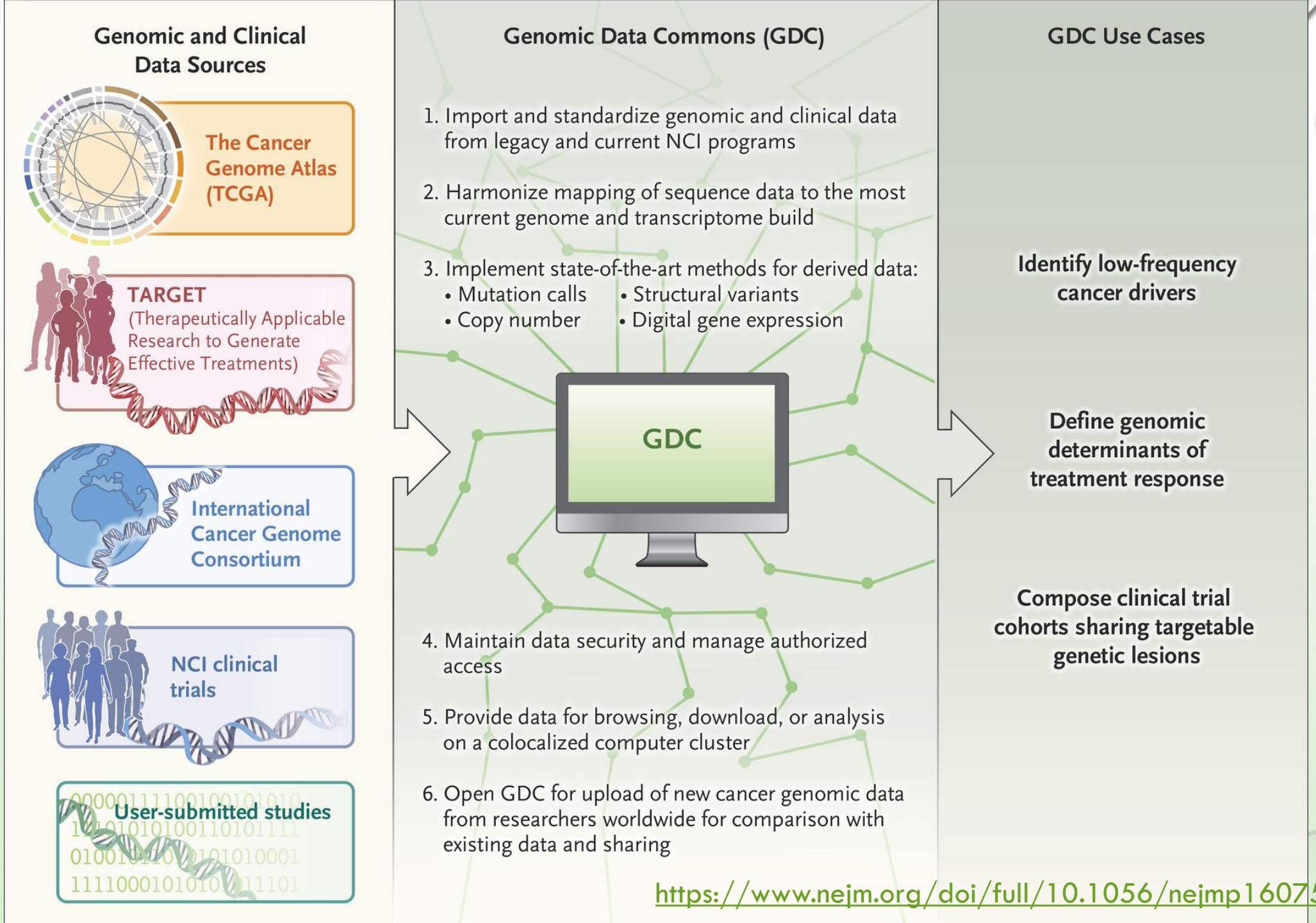
FRANCESCA M. BUFFA

WHAT IS BIOMEDICAL RAW DATA?

- SEQUENCING DATA
- MICROARRAY DATA
- PROTEIN STRUCTURE DATA
- MORPHOLOGICAL DATA
- IMAGES FROM MICROSCOPES
- CLINICAL IMAGING
- CLINICAL DATA
- ECOLOGICAL DATA
- BIOGEOGRAPHICAL DATA
- DEMOGRAPHIC DATA
- ….

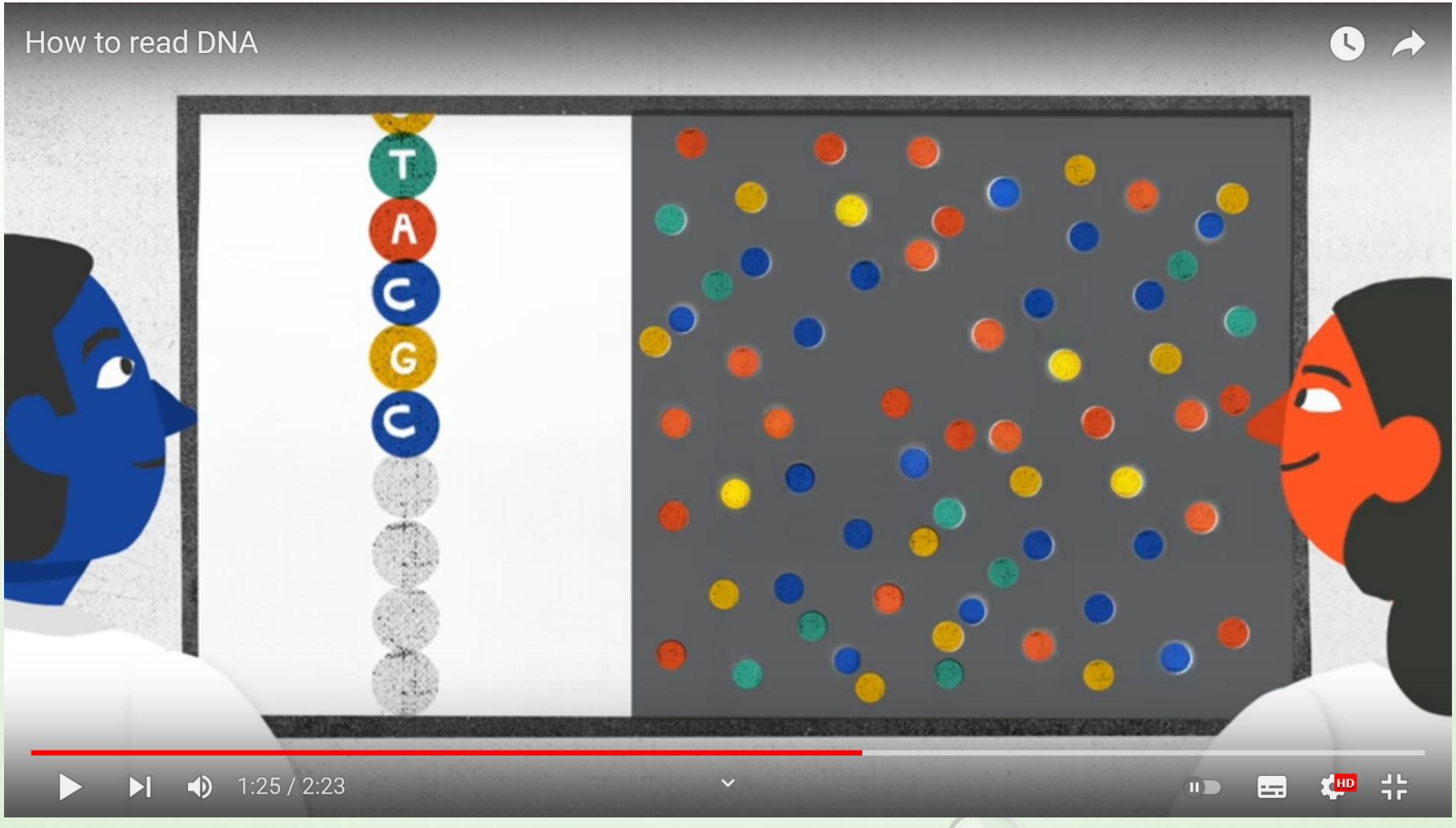


https://www.rcpath.org/profession/publications/college-bulletin/april-2021/the-100-000-genomes-project-research-and-clinical-legacy.html



https://www.nejm.org/doi/full/10.1056/nejmp1607591

DNA: WE CAN READ IT AND WRITE IT

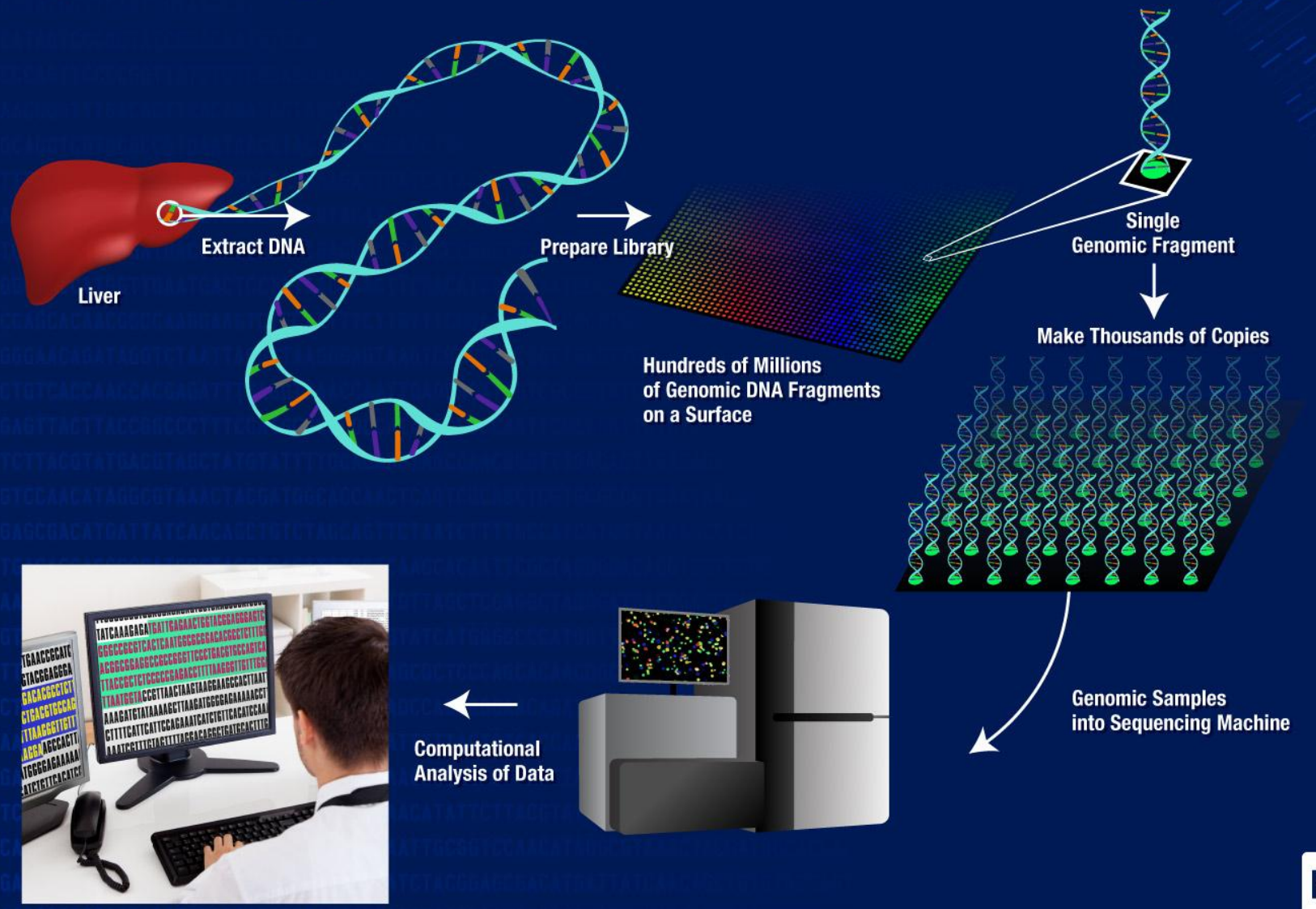


OxfordSparks

https://www.youtube.com/watch?v=bQwTa8Rf31A

Dna Sequencing

NHGRI FACT SHEETS
genome.gov

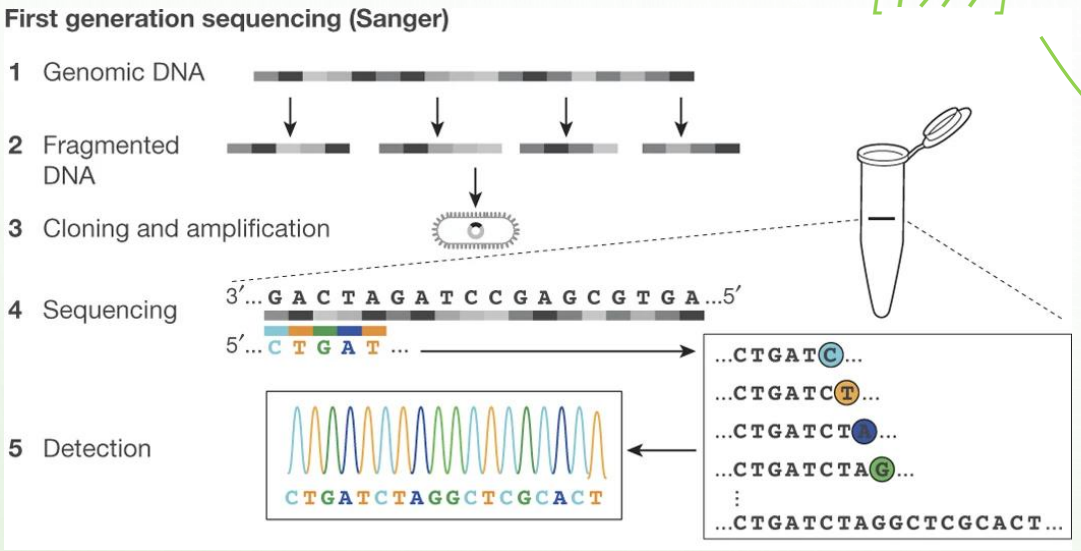


NIH National Human Genome Research Institute

https://www.genome.gov/about-genomics/fact-sheets/DNA-Sequencing-Fact-Sheet

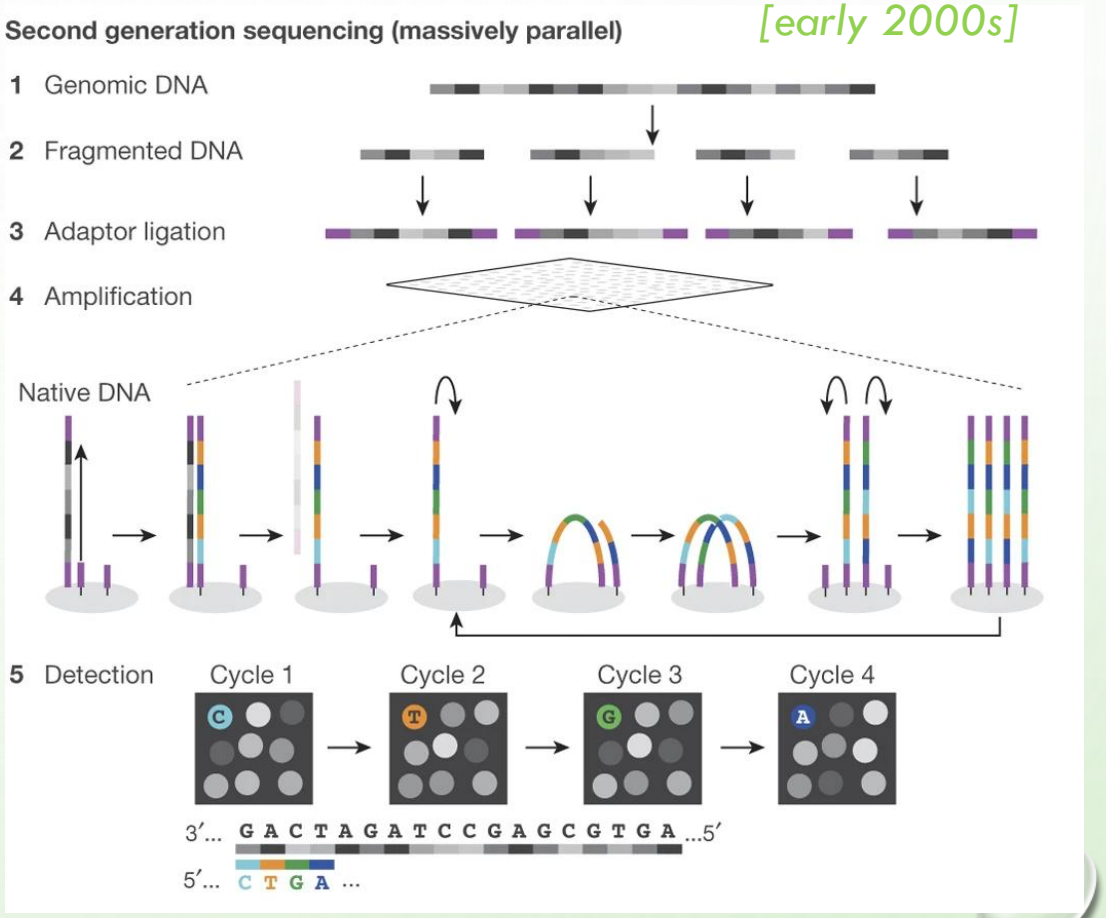
DNA SEQUENCING TECHNOLOGIES

[1977]

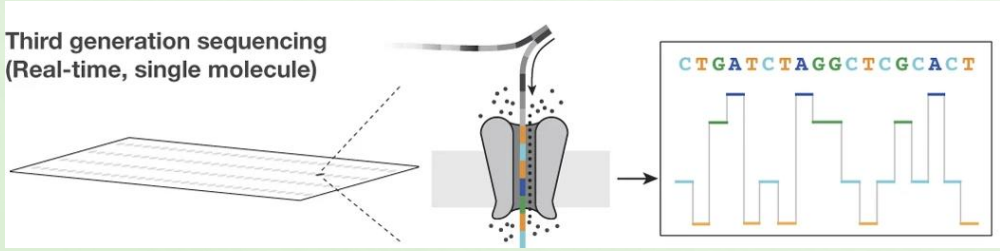


https://www.youtube.com/watch?v=ONGdehkB8jU

[early 2000s]



https://www.youtube.com/watch?v=v10bUR2aL5g



https://www.nature.com/articles/nature24286/

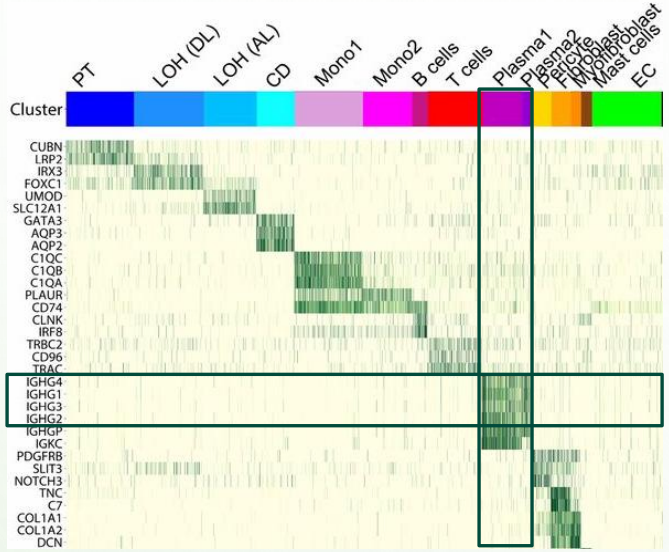
DNA/RNA STRING REPRESENTATION

```
Sequence: GTTCTCTTCTTCCCTAGCGGATAACAGATTCGGCCCACTGTCCCCTTCAG

[1] + Sequence Length: 50

[2] + Nucleotide Frequency: {'A': 8, 'C': 18, 'G': 9, 'T': 15}

[3] + DNA/RNA Transcription: GUUCUCUUCUUCCCUAGCGGAUAACAGAUUCGGCCCACUGUCCCCUUCAG

[4] + DNA String + Complement + Reverse Complement:
5' GTTCTCTTCTTCCCTAGCGGATAACAGATTCGGCCCACTGTCCCCTTCAG 3'
   ||||||||||||||||||||||||||||||||||||||||||||||||||
3' CAAGAGAAGAAGGGATCGCCTATTGTCTAAGCCGGGTGACAGGGGAAGTC 5' [Complement]
5' CTGAAGGGGACAGTGGGCCGAATCTGTTATCCGCTAGGGAAGAAGAGAAC 3' [Rev. Complement]
```

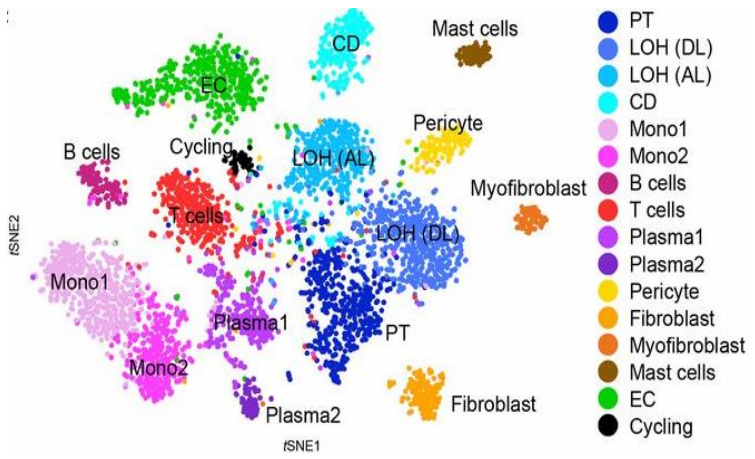
https://python.plainenglish.io/bioinformatics-in-python-dna-toolkit-part-2-transcription-and-reverse-complement-1ed194e0d06e

We can sequence RNA in single cells

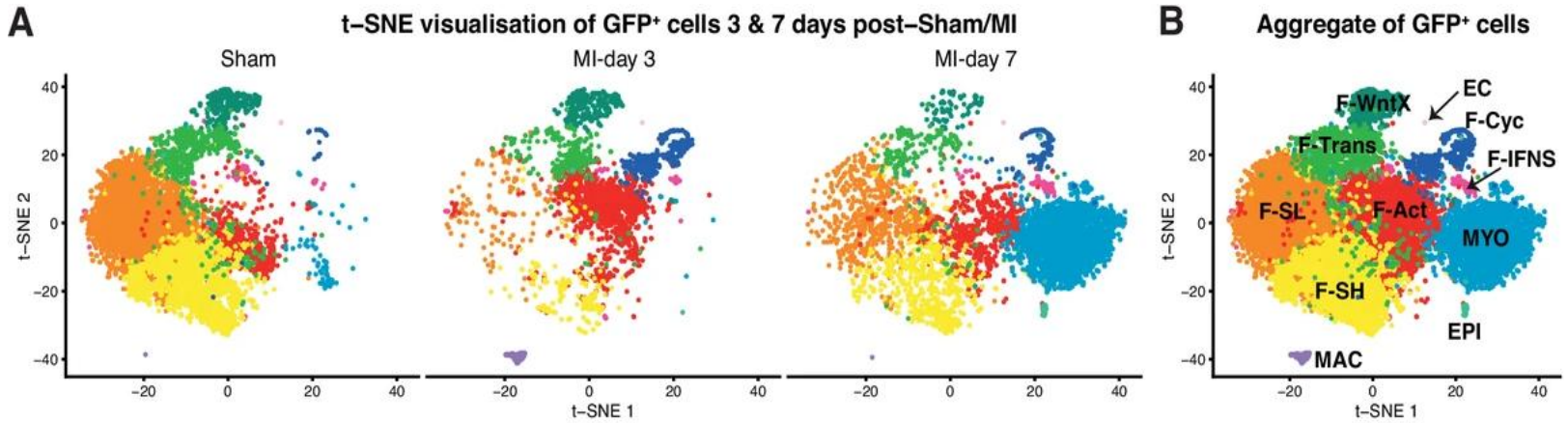
Identifying cell-type marker genes



Discovering sample heterogeneity

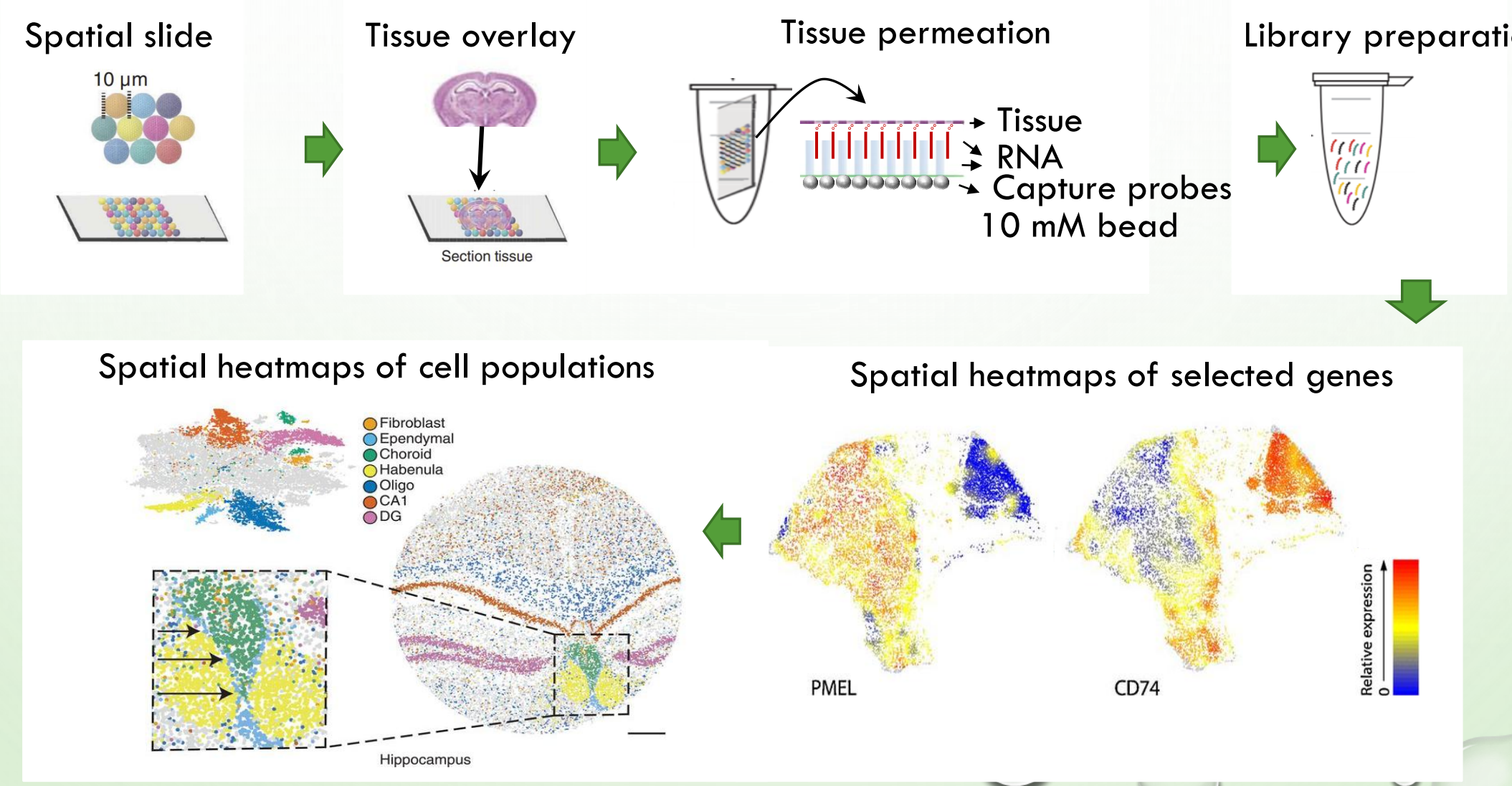


Uncovering tissue dynamics



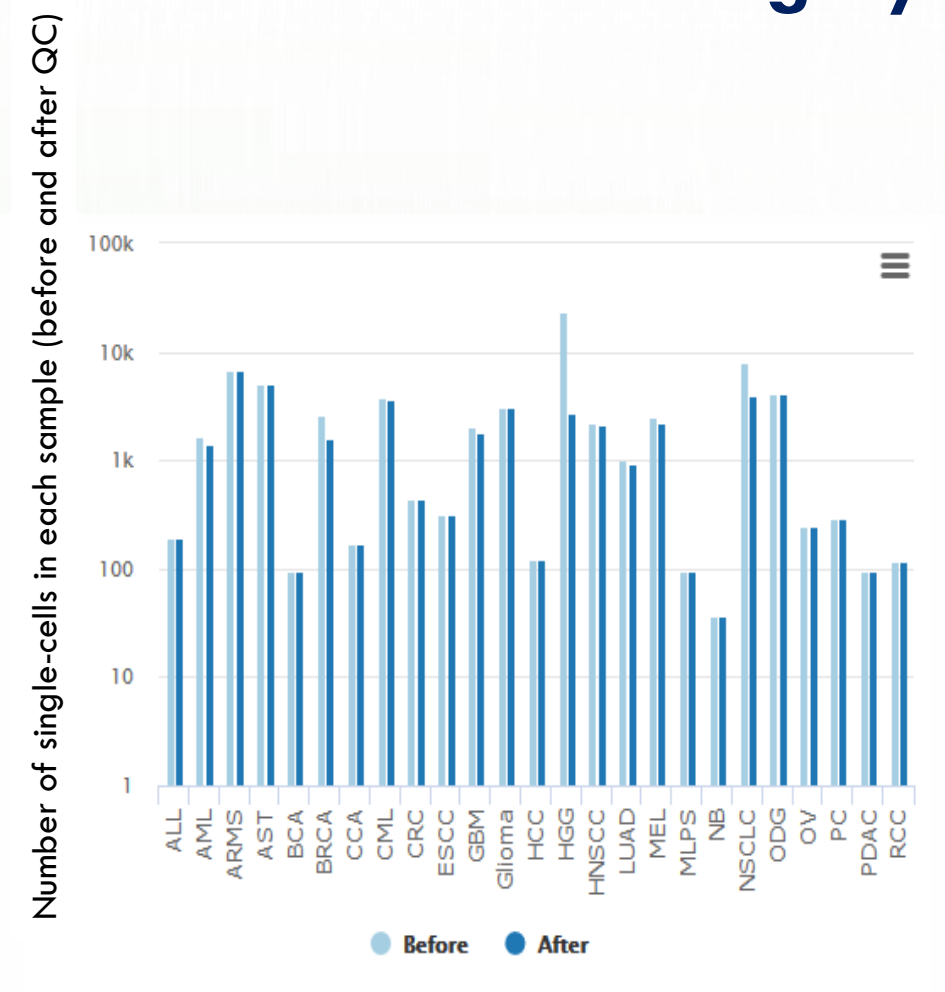
Images from: Wu et al (2018) JASN and Farbehi et al (2019) Elife

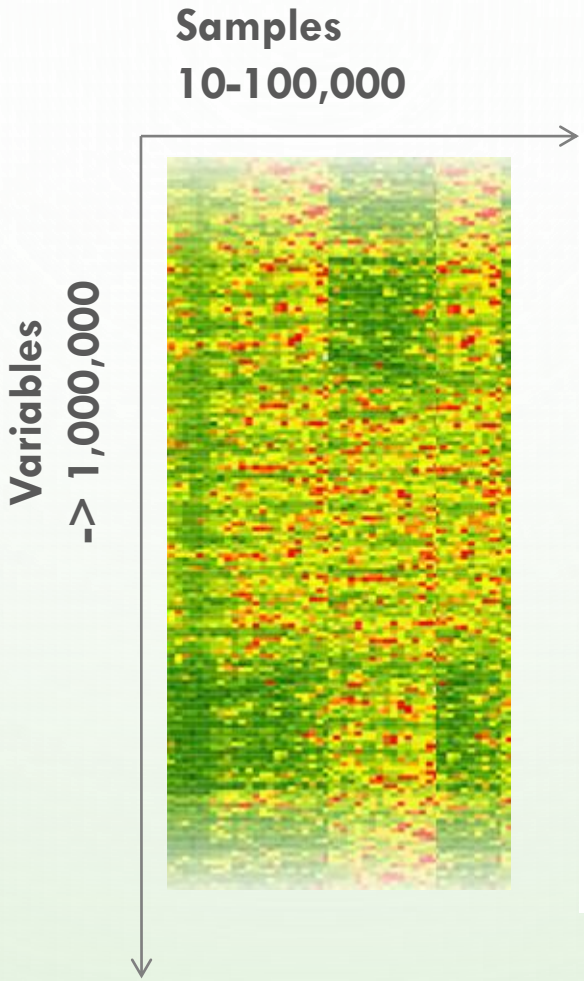
We can acquire a spatial image of single-cell expression

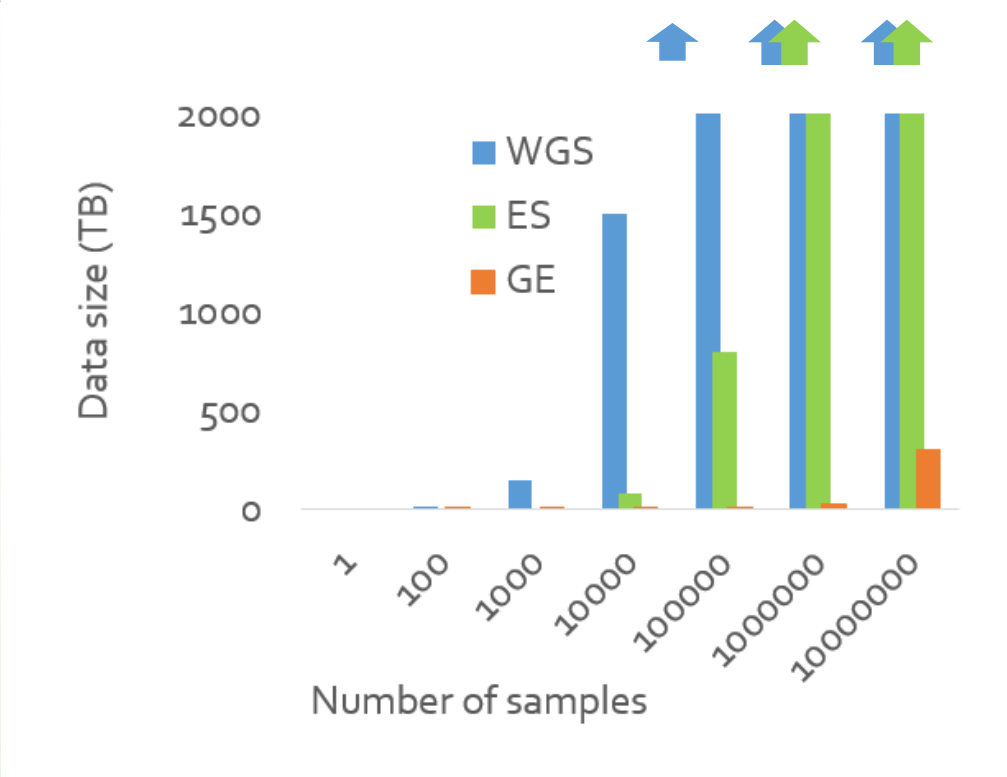


Images from: Vickovic et al (2019) Nature Methods, Thrane (2018) Cancer Res and Rodriques et al (2019) Science

Highly multidimentional datasets







http://biocc.hrbmu.edu.cn/CancerSEA/home.jsp

DNA SEQUENCING DATA

Base Calling
(Vendor tool, non-text)

FASTQ files
Sequencing Reads

FASTQ

De facto standard for storing high-throughput sequencing data

- ☐ Text-based file containing raw + quality
- ☐ Can contain millions of entries (size ~GBs)

```
@ML-P2-14:9:000H003HG:1:11102:17290:1073 1:N:0:TCCTGAGC+GCGATCTA
TTTGGTAACAGCATGAATTATTCTAGCCACTAAAACTCTATGAACATCTTGTGAAGGTTTCAGATAGAGCCTGAAGTACACAGAGAACAATTCTTAAAAAA
+
AAAAAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEAEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEEE<AEEEEEEE
```

1) Begins with @, followed by sequence identifier and optional additional information/description
2) The sequence: the base calls A, C, T, G and N
3) A separator, which is simply a plus (+) sign.
4) The base call quality scores. It represents the probability of an error in base call. These are Phred +33 encoded, using ASCII characters to represent the numerical quality scores.

DNA SEQUENCING DATA

Base Calling
(Vendor tool, non-text)

FASTQ files
Sequencing Reads

Alignment to a reference genome
or assembly of new genome

SAM/BAM files:
Aligned sequencing reads
.sam: uncompressed text file
.bam: compressed text file

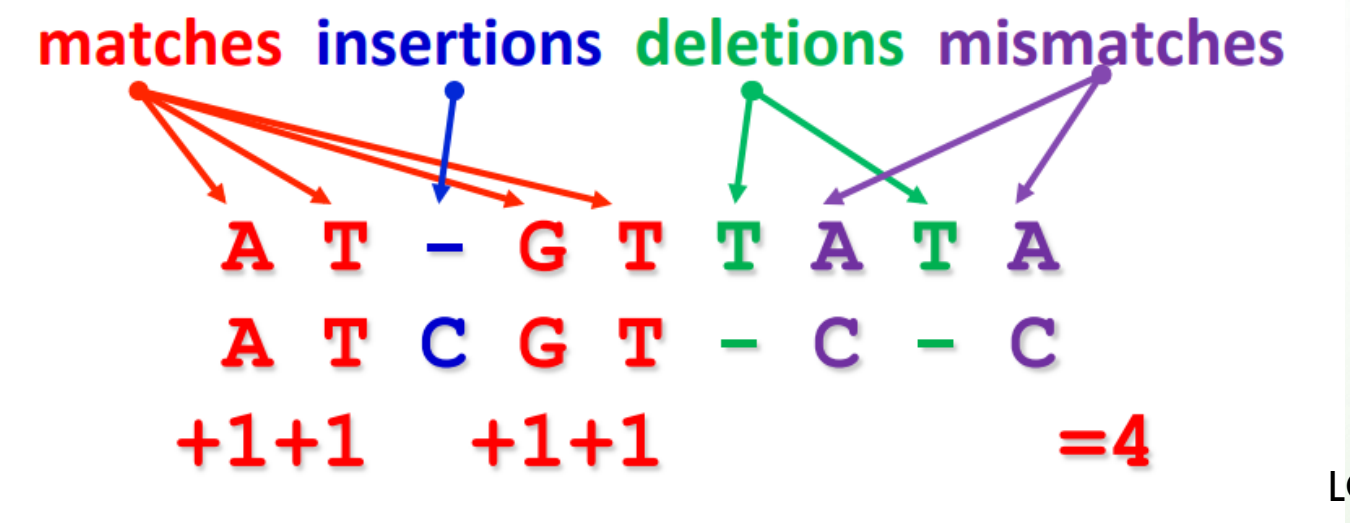
ALIGNMENT: SEQUENCE MATCHING PROBLEM

A T G T T A T A
A T C G T C C

Finding regions of similarities and dissimilarities between sequences, and infer a measure of relatedness, is vital for phylogenetic research:
Comparison of genetic material across organisms allows for example to:

- Infer functional relationships
- Infer evolutionary relationships

ALIGNMENT: SEQUENCE MATCHING PROBLEM



LONGEST COMMON SEQUENCE

Alignment of two sequences is a two-row matrix:

1st row: symbols of the 1st sequence (in order) interspersed by "-"

2nd row: symbols of the 2nd sequence (in order) interspersed by "-"

LONGEST COMMON SUBSEQUENCE

```
A T - G T T A T A
A T C G T - C - C
```

Matches in alignment of two sequences (ATGT) form their Common Subsequence

Longest Common Subsequence Problem:

Find a longest common subsequence of two strings.

- Input: Two strings.
- Output: A longest common subsequence of these strings.

For arbitrary number of input sequences, the problem is NP-hard.

When the number of sequences is constant, the problem is solvable in polynomial time with a number of approaches

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5425171/

SEQUENCE ALIGNMENT MAP - SAM FILE

TAB-delimited text format for storing alignment sequences against a reference

```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

Binary Alignment/Map (BAM) file (*.bam) is a compressed binary SAM file (smaller size + faster access)

SEQUENCE ALIGNMENT MAP - SAM FILE

TAB-delimited text format for storing alignment sequences against a reference

```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

File-level metadata. Optional.

Binary Alignment/Map (BAM) file (*.bam) is a compressed binary SAM file (smaller size + faster access)

SEQUENCE ALIGNMENT MAP - SAM FILE

TAB-delimited text format for storing alignment sequences against a reference

Reference sequence dictionary

```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
```

accepted_hits.sam

Binary Alignment/Map (BAM) file (*.bam) is a compressed binary SAM file (smaller size + faster access)

SEQUENCE ALIGNMENT MAP - SAM FILE

TAB-delimited text format for storing alignment sequences against a reference

```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
```

accepted_hits.sam

Program

Binary Alignment/Map (BAM) file (*.bam) is a compressed binary SAM file (smaller size + faster access)

SEQUENCE ALIGNMENT MAP - SAM FILE

TAB-delimited text format for storing alignment sequences against a reference

Alignment information

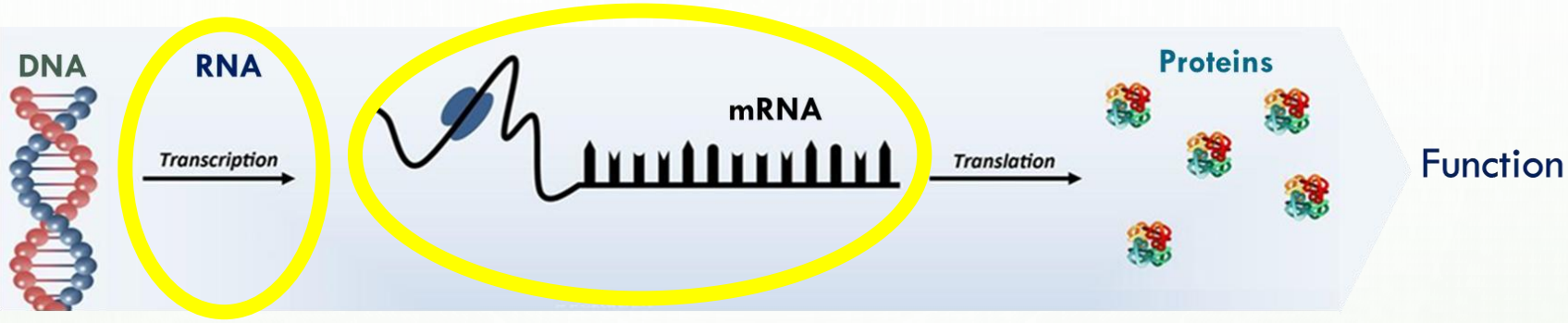
```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714       16      chr20   190930  3       100M    *       0       0
       CCGTGTTTAAAGGTGGATGCGGTCACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCCTCT
C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961       16      chr20   193953  50      100M    *       0       0
       TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030       16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699       0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
```

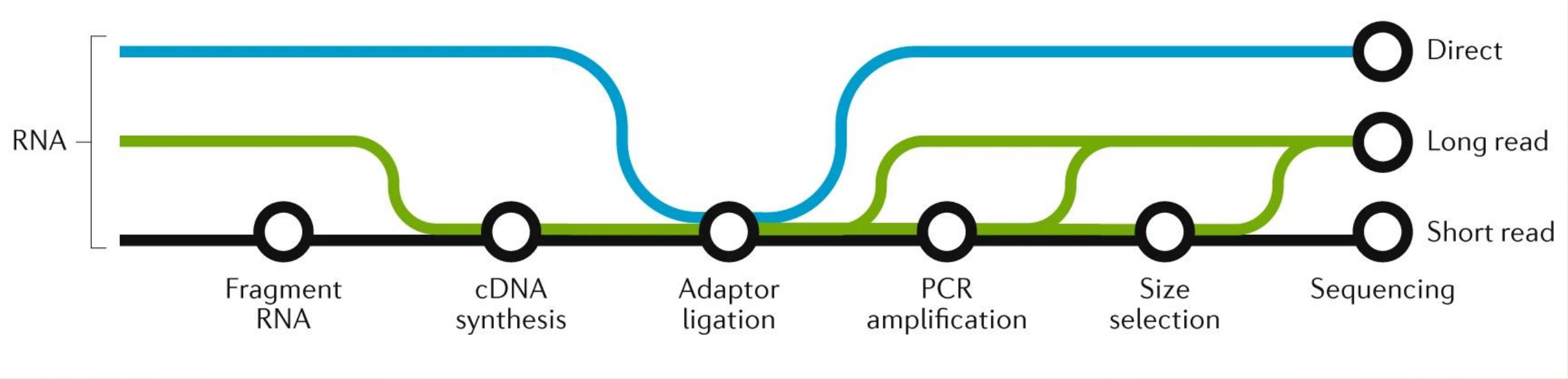
accepted_hits.sam

http://samtools.github.io/hts-specs/SAMv1.pdf

Binary Alignment/Map (BAM) file (*.bam) is a compressed binary SAM file (smaller size + faster access)

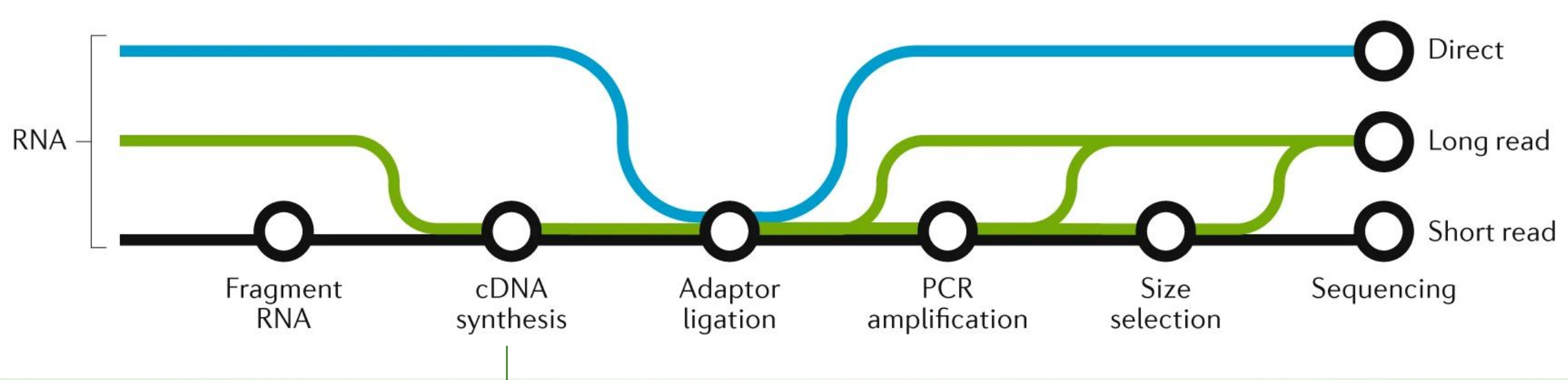
RNA SEQUENCING





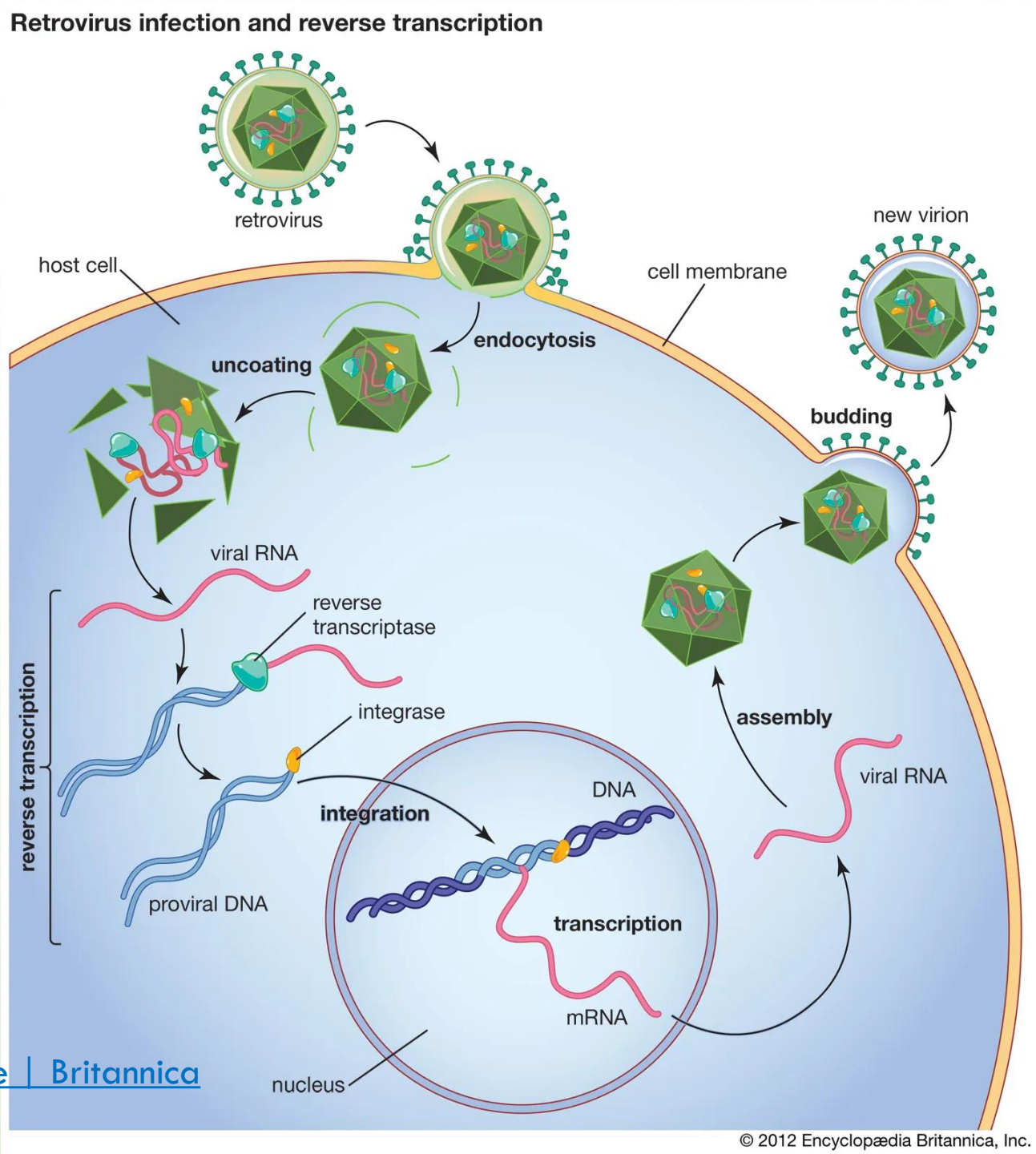
https://www.nature.com/articles/s41576-019-0150-2/figures/1

RNA SEQUENCING



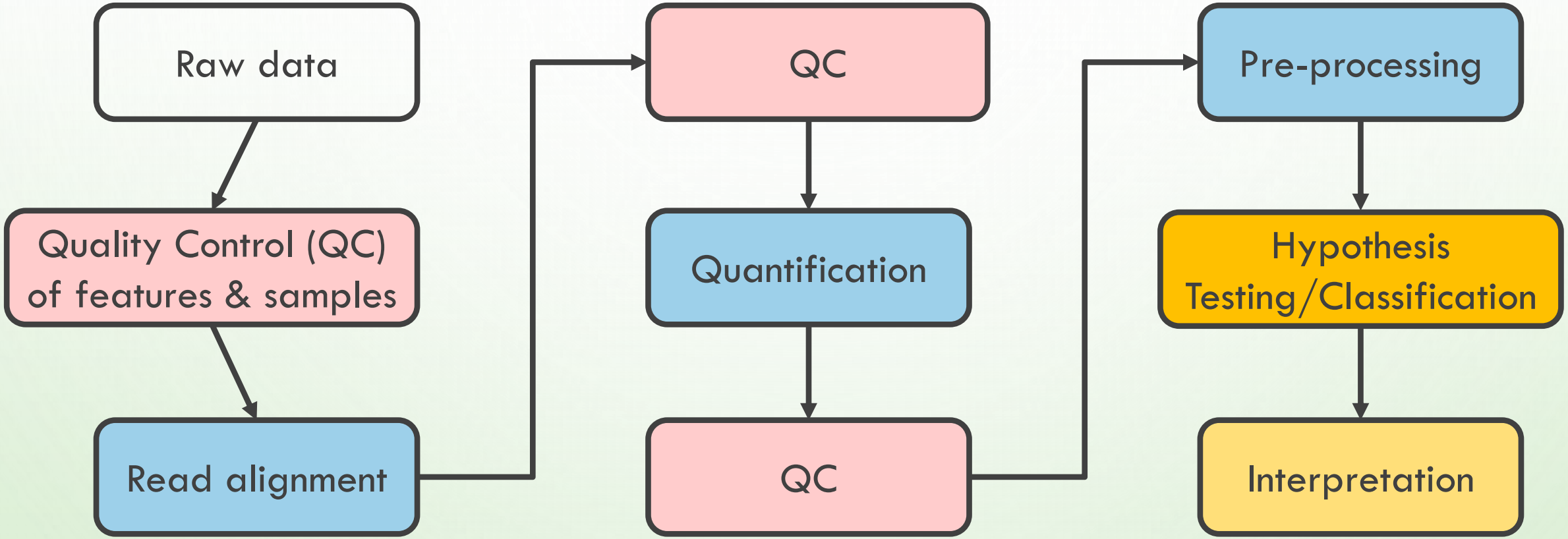
NOTE: Complementary DNA (cDNA) is generated via reverse transcription: mRNA is the template and DNA is the product

https://www.nature.com/articles/s41576-019-0150-2/figures/1

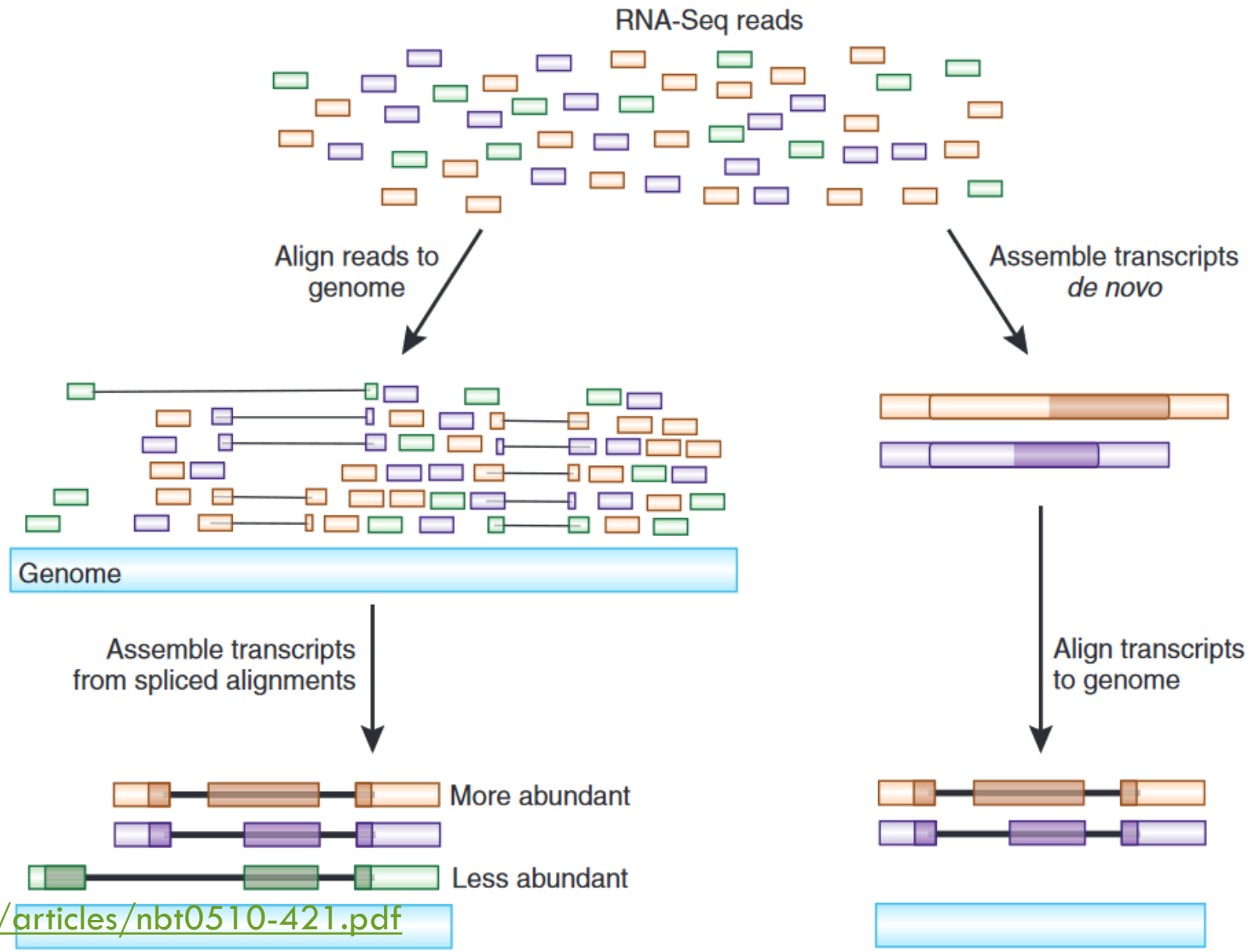


Encyclopedia Britannica
reverse transcriptase | enzyme | Britannica

RNA-SEQ: FROM LOOKING AT THE DATA TO ANALYSIS



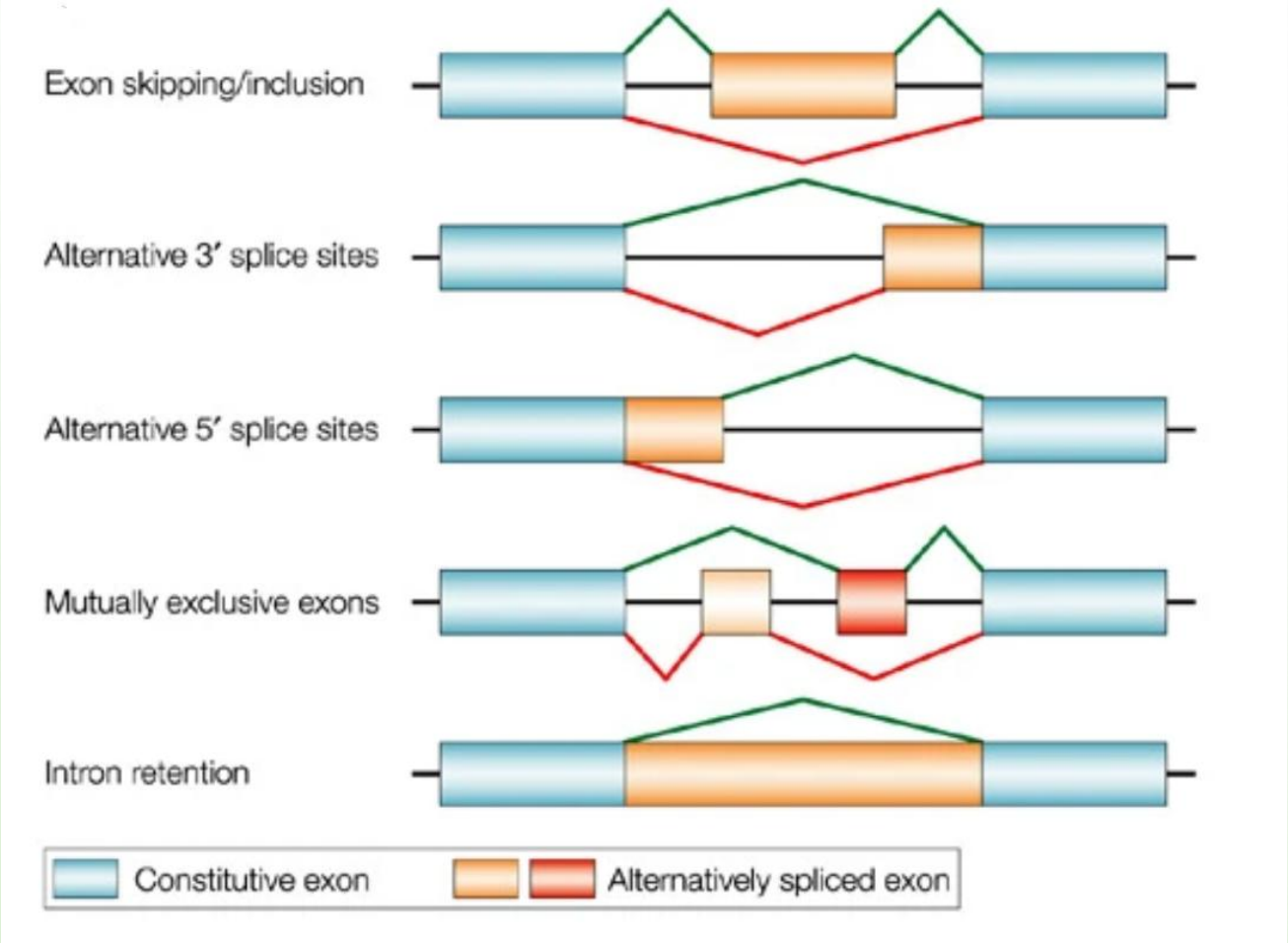
RNA-SEQ ALIGNEMENT

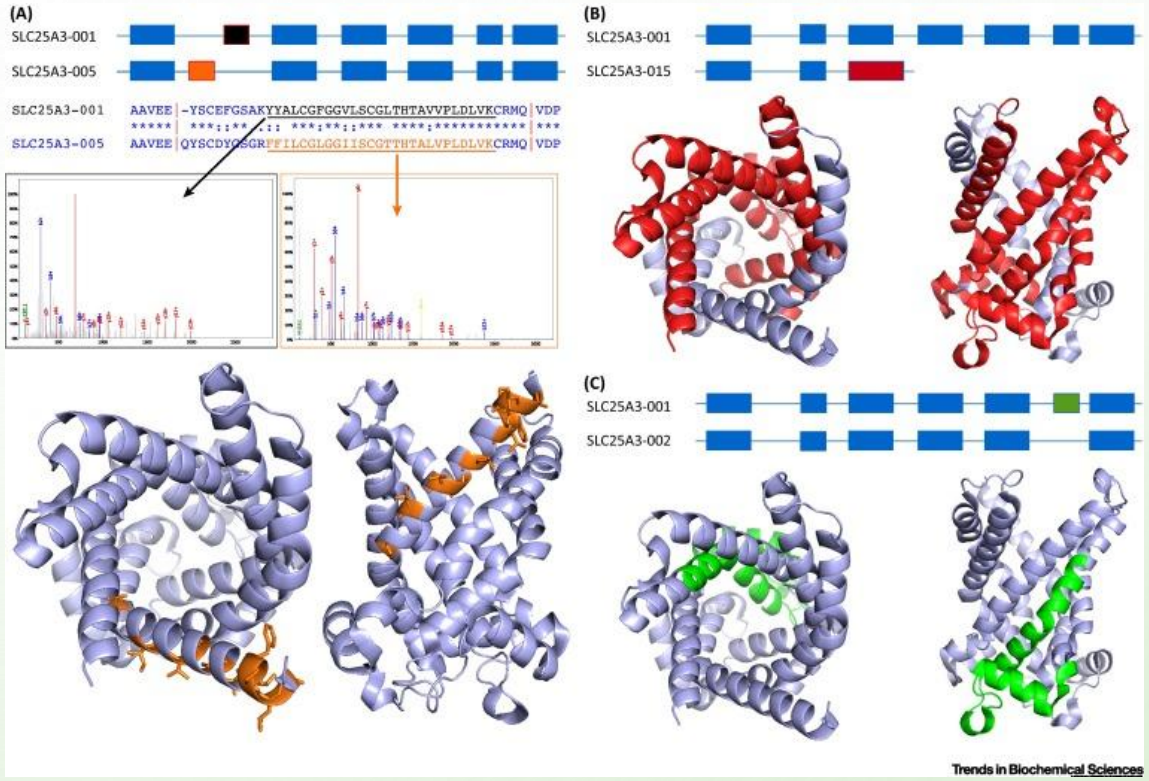


Haas & Zody, 2010
https://www.nature.com/articles/nbt0510-421.pdf

HOW CAN SO FEW GENES ENCODE FOR THE COMPLEXITY OF EUKARYOTES?

From one RNA many possibilities for proteins





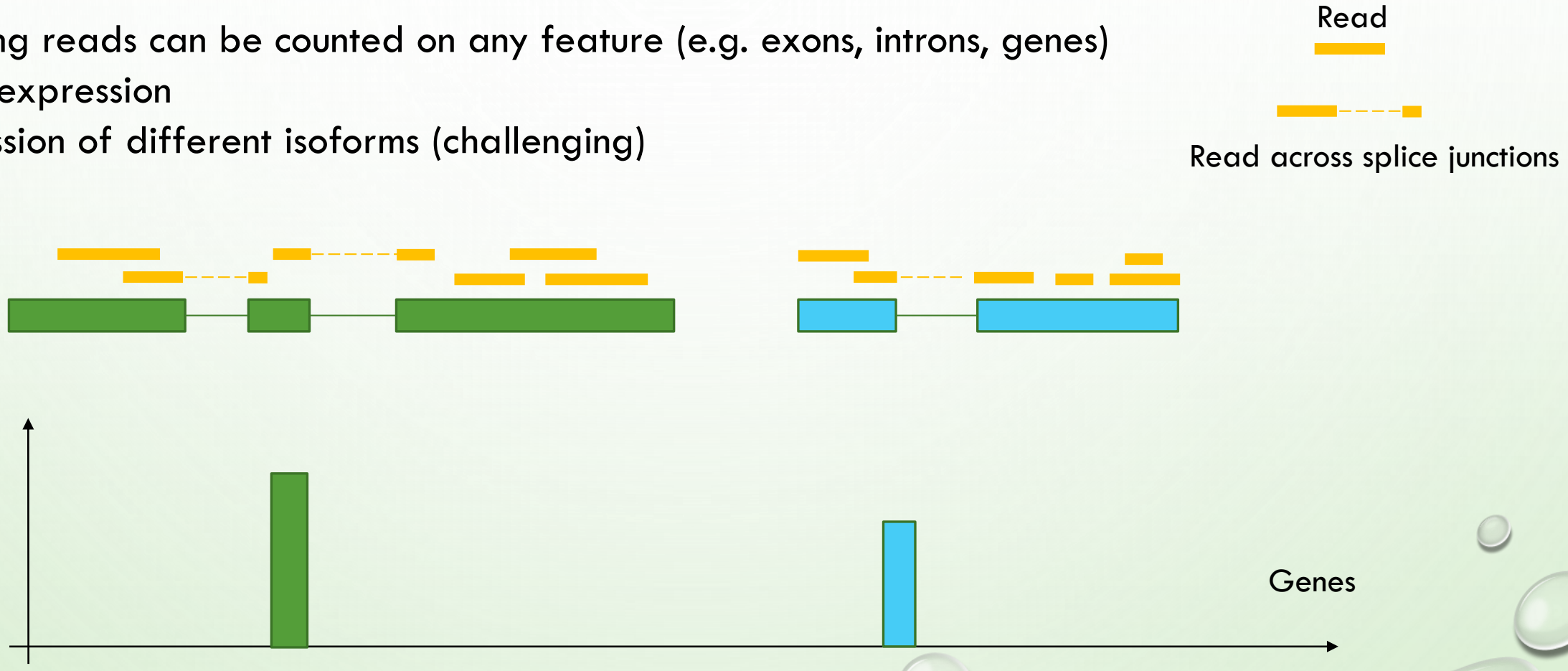
https://www.nature.com/articles/nrg775

https://www.cell.com/trends/biochemical-sciences/fulltext/S0968-0004(16)30118-9

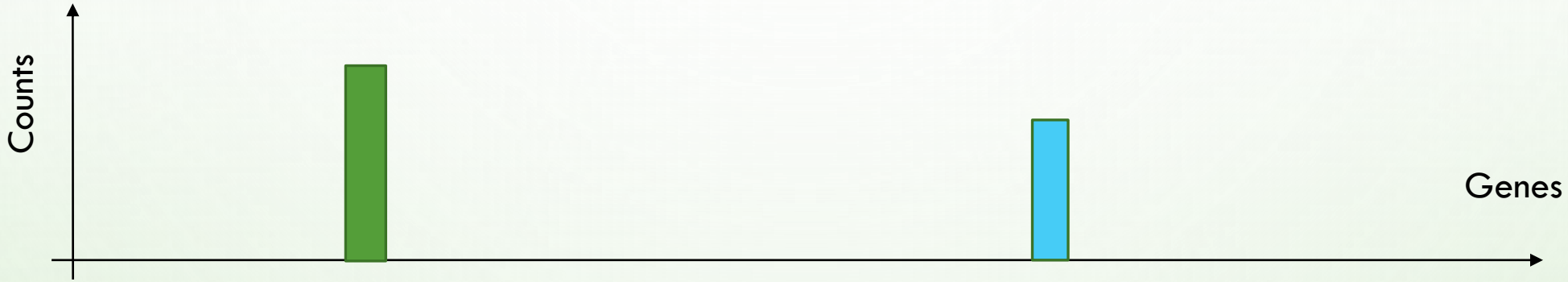
QUANTIFYING GENE EXPRESSION

Sequencing reads can be counted on any feature (e.g. exons, introns, genes)

- Gene expression
- Expression of different isoforms (challenging)

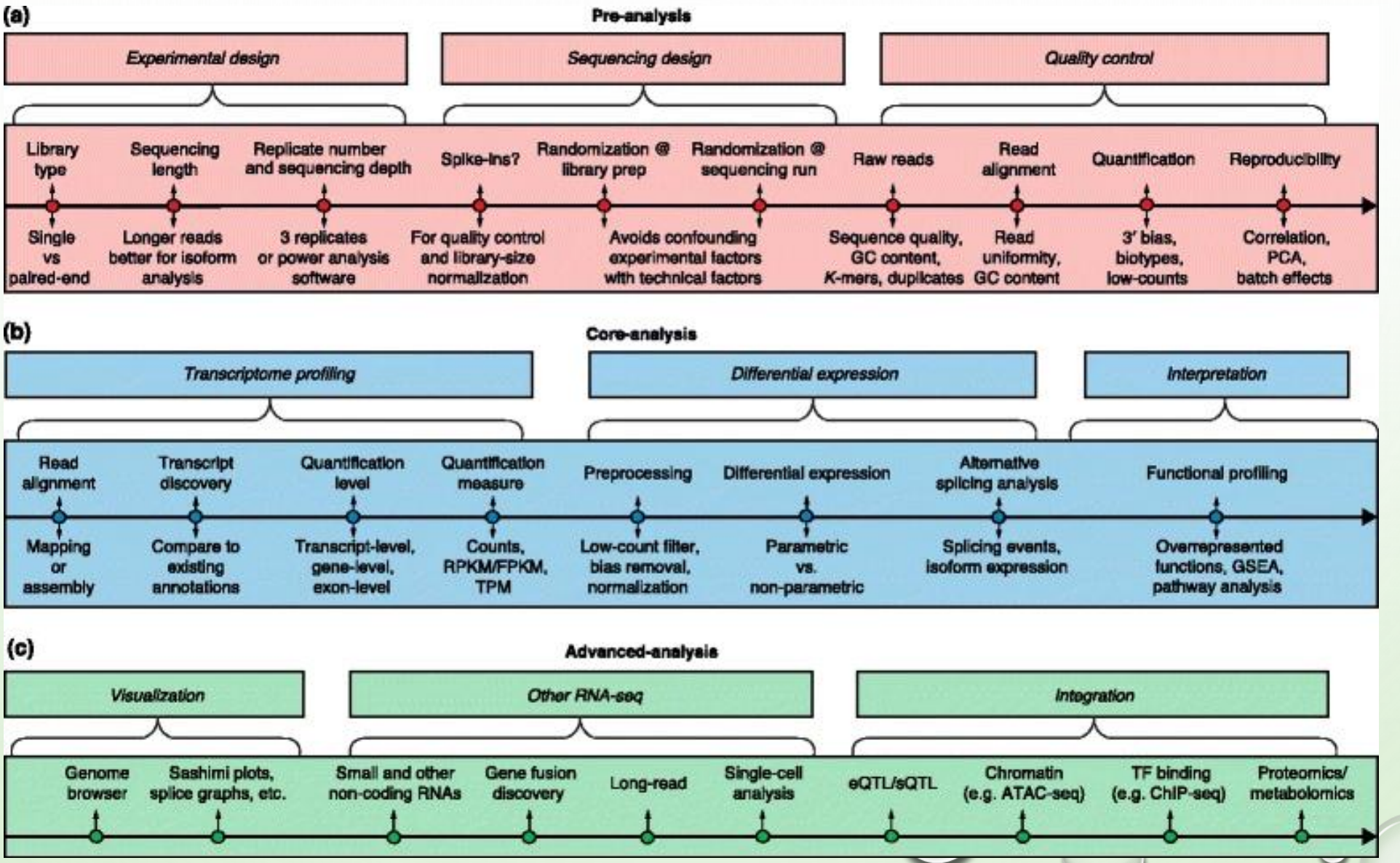


COUNTS/GENE EXPRESSION MATRIX



	Sample1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7
Gene A	345	0	23	56	76	4	3
Gene B	60	45	56	32	24	58	54
Gene C	0	0	0	0	0	0	0
Gene D	453	569	764	897	564	432	865

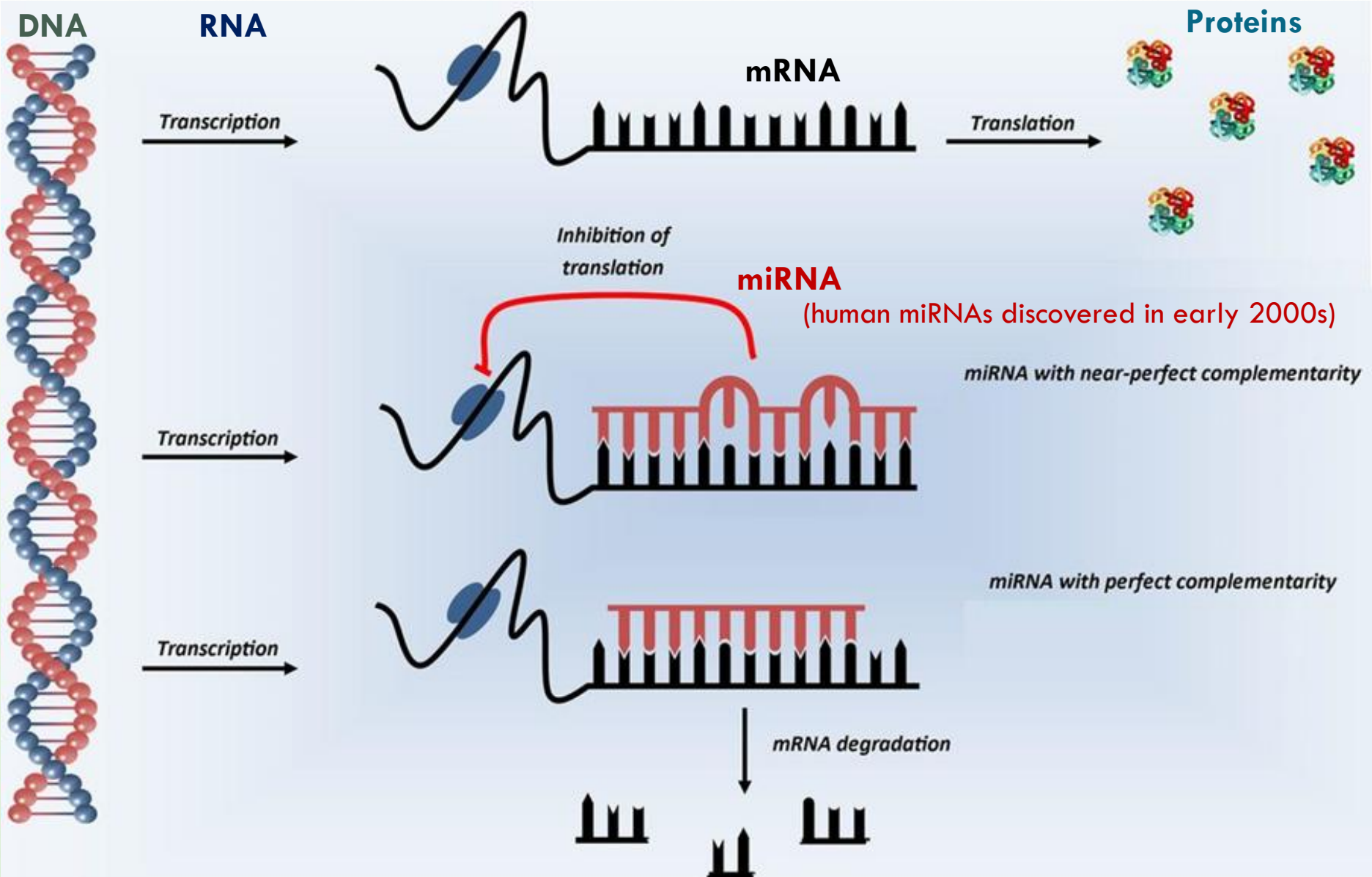
RNA-SEQ DATA ANALYSIS



https://genomebiology.biomedcentral.com/articles/10.1186/s13059-016-0881-8

HOW CAN SO FEW GENES ENCODE FOR THE COMPLEXITY OF EUKARYOTES?

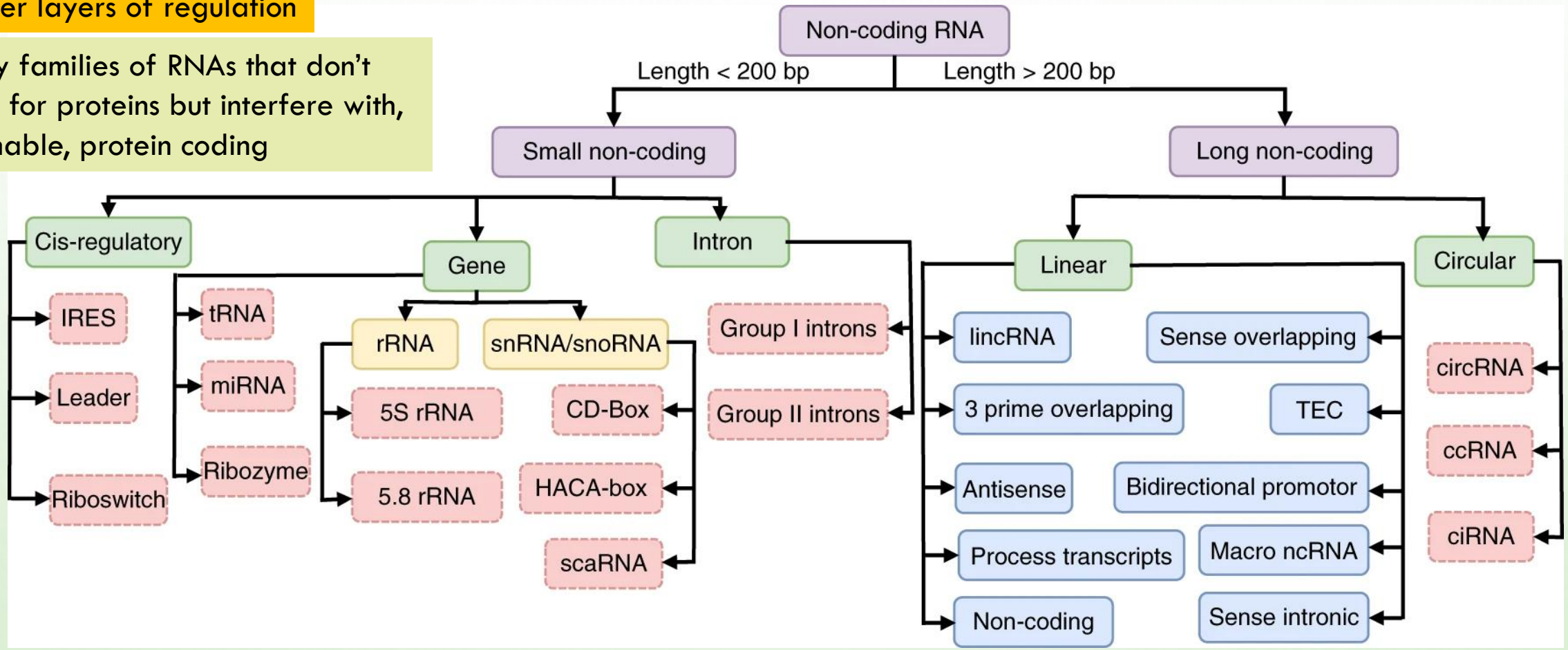
Further layers of regulation



HOW CAN SO FEW GENES ENCODE FOR THE COMPLEXITY OF EUKARYOTES?

Further layers of regulation

Many families of RNAs that don't code for proteins but interfere with, or enable, protein coding



https://www.nature.com/articles/s42256-019-0051-2